

# 32A — Module 32 Novelty Audit and Contribution Boundary

Professional reading edition of the evolving public MKUFT canon

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| <b>Scientific affiliation</b> | Independent Researcher                                                        |
| <b>Object status</b>          | Research module reading edition — not a standalone Zenodo publication         |
| <b>Public source</b>          | <a href="#">Open canonical public source</a>                                  |
| <b>Source commit</b>          | 26e06b6fea52e6b052dbdd540e80d1d926b96b07                                      |
| <b>MKUFT backbone DOI</b>     | <a href="https://doi.org/10.5281/zenodo.17780566">10.5281/zenodo.17780566</a> |

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# 1. Purpose

Module 32 began from an independently reached chain about recursive learning, self-maintained constraint, changing future reach, and scale-dependent freedom/capability. Same-day refinement then added load-bearing-set tests, relation-versus-carrier substitution, relational addressability, same-self parity, temporal-continuity kernels, typed relation scopes, completion fibers, quotient-based scale readout, macrostate realisation classes, temporal mismatch exposure, an  $I \rightarrow P$  transition-support assay, an intrinsic/extrinsic address split with path transport, loop-history tests, context-indexed P-realisation fibers and boundary-conditioned physical comparators, a capstone factorisation in which the complete addressed state/context produces a typed admissibility descriptor and active physical law object, and finally a dynamic-interface promotion gate specifying when a load-bearing relation must be represented explicitly in the recursively updated state.

A literature audit found substantial prior and adjacent work. The scientific task is therefore not to defend every ingredient as new. It is to identify what prior work already supplies, what the current MKUFT integration adds operationally, and what remains only a candidate contribution pending broader review and empirical use.

The governing rules are:

*Independent convergence can support confidence in a reasoning route, but scientific novelty belongs only to the residual structure not already supplied by prior work.*

*A new canon write is a changed object. Re-audit the integrated object; if the write exposes another load-bearing relation or contradiction, continue refinement rather than treating the successful commit as closure.*

The second rule is process provenance, not a novelty claim about scientific method.

## 2. Strong prior art that removes broad novelty claims

### 2.1 Constraint closure and autonomous organisation

Autopoiesis, autonomy/enaction, and closure-of-constraints work already treat living organisation as relational, self-maintaining, boundary-dependent, historically structured, and coupled to its environment. Montévil and Mossio (2015) explicitly formalise biological organisation as closure of constraints.

**Boundary:** MKUFT does not claim invention of constraint closure, autopoiesis, relational individuality, viability, self-maintenance, or organism–environment coupling.

### 2.2 Changing possibility spaces and accessible-state topology

Longo, Montévil and Kauffman (2012), Kauffman and Roli (2021), and Adams et al. (2024) develop changing biological possibility spaces, internally maintained constraints, accessible-state structure, and emergent/local rule change.

**Boundary:** MKUFT does not claim invention of changing phase/possibility spaces, enablement, internally maintained constraints, or the broad claim that organisation can alter accessible states.

### 2.3 Semantic/symbol–matter control

Pattee and related biosemiotic work already study symbol–matter, semantic closure, coding, measurement, and physical constraint.

**Boundary:** MKUFT does not claim invention of semantic closure or the general claim that informational organisation can be physically realised through constraints.

### 2.4 Ablation, multiple realisation, coarse-graining, fibers and quotients

Ablation, necessary-condition tests, alternative physical realisation, quotient descriptions, fibers/preimages, equivalence classes, coarse-graining, macrovariables, and multiple realisability are established mathematical and scientific methods.

Hoel, Albantakis and Tononi (2013) explicitly formalise cases in which coarse-grained macrostates have many microstate realisations and can exhibit greater effective causal information than the microscopic description.

**Boundary:** 32S/32S3 do not claim invention of these ingredients or the general idea that one macrostate can have many microstate realisations.

## 2.5 Local-to-global mathematics and contextual covers

Nerve constructions, sheaves, restriction/gluing, and local-to-global consistency are established mathematics. Robinson (2018) develops approximate consistency through a consistency radius for sheaves of pseudometric spaces. Abramsky and Brandenburger (2011) use sheaf theory over measurement contexts to formulate non-locality and contextuality.

**Boundary:** 32S3 does not claim invention of nerve complexes, sheaves, consistency residuals, contextual covers, gluing, or local-to-global obstruction.

The Abramsky–Brandenburger case is also a useful hygiene reminder: a local context in local-to-global mathematics need not mean a literal spatially adjacent patch.

## 2.6 Hypergraphs and higher-order interactions

Hypergraphs, simplicial complexes, and higher-order networks are established tools for relations involving more than two components. Zhang, Lucas and Battiston (2023) show that the chosen higher-order representation can itself change collective dynamical predictions.

**Boundary:** MKUFT does not claim invention of arbitrary higher-order relation scopes or hypergraph-style representation. 32S3 uses typed relation scopes because a purely geometric nerve would be too restrictive for the framework's own stated relation classes.

## 2.7 Causal semantic information

Kolchinsky and Wolpert (2018) define semantic information using counterfactual scrambling and causal contribution to system persistence.

**Boundary:** 32S3 does not claim invention of intervention-based semantics, causal scrambling, or the idea that meaning can be operationalised by consequence.

## 2.8 Information conversion, synergy, and causal emergence

Causal-emergence and partial-information work already formalise changes in informational organisation across scales. Varley and Hoel (2022) show coarse-graining can convert redundant, unique, and synergistic information, including increased synergy at a macroscale.

**Boundary:** MKUFT does not claim invention of macro-scale synergy or the broad idea that a macrodescription may expose information structure obscured at a microdescription.

## 2.9 Holographic phase encoding and reconstruction

Optical and metasurface holography are established physics. Huang et al. (2013) demonstrate 3D holographic reconstruction with subwavelength elements whose orientations encode phase.

**Boundary:** holography is a physical comparator and deformation template only. MKUFT does not claim invention of phase encoding, Fourier/Fresnel reconstruction, distributed optical information, or a universal holographic principle for biology or reality.

## 2.10 P-layer biological comparators remain P-layer evidence

Guruciaga et al. (2026) show boundary geometry controlling topological-defect configuration and lumen nucleation location in mouse embryonic tissue. Schone et al. (2025) show stable cortical hand/finger representations after arm amputation.

**Boundary:** these are useful P-layer comparators for relational address, boundary-conditioned interior structure, and persistence after endpoint removal. Neither establishes an independent I layer or  $I \rightarrow P$  mechanism.

## 2.11 Connections, parallel transport, holonomy, and geometric phase are prior work

Fiber bundles, connections, parallel transport, and holonomy are standard differential geometry and mathematical physics. Simon (1983) identified Berry's geometric phase with holonomy in a Hermitian line bundle, and Berry (1984) derived the path-dependent geometric phase acquired by an adiabatically transported quantum eigenstate.

**Boundary:** 32S4 does not claim invention of intrinsic/extrinsic decomposition, parallel transport, holonomy, geometric phase, or the fact that equal endpoints can retain path-dependent state information in systems where the relevant connection exists.

Its use is methodological: make path transport explicit inside the existing addressed-state, identity-history, and admissibility architecture.

## 2.12 Boundary-conditioned field spectra, Casimir geometry, and driven vacuum amplification are prior work

Boundary-value problems, cavity modes, Casimir interactions, fluctuation-induced forces, and the dynamical Casimir effect are established physics. Nanostructured Casimir experiments show measurable geometry/material dependence;

Wilson et al. (2011) demonstrated real photon generation under rapid driven modulation of an effective electromagnetic boundary.

Jaffe (2005) also provides an important interpretive guard: Casimir effects need not be treated as proof of a literal extractable zero-point-energy reservoir.

**Boundary:** 32S4 does not claim invention of boundary-conditioned spectra, Casimir geometry, dynamical Casimir photon generation, vacuum fluctuations, or quantum passivity. These are P-layer comparators and energy-accounting constraints, not evidence for independent I→P dynamics or free energy.

## 2.13 Constrained law domains, conditional dynamics, measurement instruments, and GR constraint surfaces are prior work

Constrained dynamical systems and viability theory already study evolution restricted by admissible sets. Probability theory and Markov-kernel/category methods already separate state spaces, processes and conditioning. Maximum Caliber and related path-ensemble methods already impose dynamical constraints on path distributions. In quantum mechanics, operator domains and boundary conditions can select self-adjoint Hamiltonian realisations and therefore admissible unitary evolution. Quantum instruments already jointly encode outcome statistics and post-measurement physical state change. General relativity already uses Hamiltonian/momentum constraint surfaces, and bounded Hamiltonian formulations require appropriate boundary terms/conditions.

**Boundary:** Module 33 does not claim invention of domain before generator, constrained dynamics, self-adjoint extensions, conditional/path measures, quantum instruments, ADM constraints or Hamiltonian boundary conditions. Nor does it claim that arbitrary relation states are known to assemble new physical laws. Its candidate contribution is the exact typed integration of those established mathematical roles with the already canonical MKUFT address/admissibility, completion, scale, transport and P-only-null machinery.

## 2.14 Junctions, dynamic interfaces, and bridge geometries are prior work

Physics and mathematics already treat boundaries, junctions, interfaces and hypersurfaces as state-bearing or dynamical objects when the governing model requires it. Israel's thin-shell formalism treats the histories of relativistic boundary surfaces geometrically, while Einstein and Rosen's 1935 construction represents two sheets joined by a bridge.

**Boundary:** Module 33S1 does not claim invention of dynamic interfaces, junction conditions, thin-shell dynamics, stateful boundaries, recursive coupled systems, or Einstein–Rosen bridge geometry. The Einstein–Rosen construction is used only as a bounded geometric comparator for the idea that a connecting structure can belong to the complete larger object; it is not evidence that  $\mathcal{J}_{AB,t}$  is a wormhole, a traversable shortcut, or an independent I-layer structure.

The residual question is operational rather than geometric: when must an already permitted relation be promoted from bundled context into explicit recursively updated state?

## 3. Component-level novelty matrix

| Module 32 family component                           | Prior-art status                                | Residual MKUFT role                                                                                  | Current novelty confidence                   |
|------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Constraint closure / self-maintenance                | Established                                     | Typed component of broader adaptive audit                                                            | Low in isolation                             |
| Changing possibility/state spaces                    | Established/close prior                         | Explicit construction address plus comparison-validity guard                                         | Low broad claim; moderate operational device |
| Endogenous constraints alter accessible states/rules | Close prior                                     | Integrates admissibility, history, maintenance and addressed reach                                   | Low principle; moderate integration          |
| No-false-subtraction/state-parity guard              | Standard mathematics requires lawful comparison | Makes comparison validity first-class in adaptive changing-address systems                           | Moderate candidate operationalisation        |
| Scale-separated local freedom vs higher capability   | Enabling constraints established                | Requires separate lower feasible-state and higher capability observables plus overconstraint control | Moderate–high candidate operationalisation   |

|                                             |                                                             |                                                                                                                 |                                                 |
|---------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Minimal load-bearing set / ablation weight  | Established methods                                         | Integrates necessity/deformation into addressed capability geometry                                             | Low ingredients; moderate integration           |
| Relation-versus-carrier substitution        | Strong antecedents                                          | Requires function/relation preservation before calling relation invariant                                       | Low-moderate ingredients; moderate integration  |
| Effective-object scale promotion            | Coarse-graining/macrovariables established                  | Promotion requires persistence, intervention, new capability, compression and boundary specificity              | Moderate integration                            |
| Relational addressability / same-self guard | Strong relational-individuality prior art                   | Separates physical persistence, independent deployment, address parity and continuity maps                      | Moderate candidate integration                  |
| Temporal identity horizon                   | Historicity/path dependence established                     | Minimum load-bearing recent history established by truncation/shuffle/held-out tests                            | Moderate candidate integration                  |
| Typed relation scopes                       | Hypergraphs/higher-order networks established               | Prevents geometric overlap from silently standing for every relation class                                      | Low mathematics; useful framework correction    |
| Compatible assignment region                | Constraint satisfaction/local-to-global methods established | Provides one typed parent object for completion, mismatch and scale readout                                     | Low mathematics; moderate integration           |
| Missing-address completion fiber            | Fibers/projections/inverse problems established             | Separates narrow completion, broad ambiguity, local incompatibility and background inconsistency                | Low mathematics; moderate integration           |
| Macrostate realisation class                | Multiple realisability/coarse-graining established          | Connects lower-level turnover to higher-level readout persistence inside Module 32 identity/scale machinery     | Low principle; moderate integration             |
| Fiber-quotient scale hand-off               | Standard fiber/quotient mathematics                         | Uses one compatibility geometry for zoom-in completion and zoom-out scale readout, with promotion kept separate | Low mathematics; moderate candidate synthesis   |
| Temporal mismatch exposure                  | Error accumulation/memory kernels established               | Keeps mismatch distinct from measured cost and requires a separate coupling test                                | Low ingredients; moderate discipline            |
| I→P transition-kernel/support assay         | Conditional kernels/causal inference established            | Tests changed P support/weighting without assuming a hidden message carrier and retains P-only null             | Low ingredients; moderate integration           |
| Intrinsic/extrinsic address split           | Standard invariance/covariance and embedding ideas          | Prevents preserved internal relation from being confused with unchanged contextual address                      | Low mathematics; moderate framework integration |
| Path/loop transport guard                   | Connections, transport and holonomy established             | Joins same-endpoint/different-path testing to MKUFT                                                             | Low ingredients; moderate integration           |

|                                                          |                                                                                            |                                                                                                                                                                                                                                                                                                                               |                                                           |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|                                                          |                                                                                            | identity/history and admissibility claims                                                                                                                                                                                                                                                                                     |                                                           |
| Context-indexed P-realisation fiber                      | Parameterised state families/fibers established                                            | Separates relational specification, compatible P-realisation and actual P-occupancy                                                                                                                                                                                                                                           | Low mathematics; moderate integration                     |
| Addressed P transition support                           | Addressed state families already owned by Module 32                                        | Corrects fixed-P shorthand so $I \rightarrow P$ support can include a changed P-construction address under comparison guards                                                                                                                                                                                                  | Moderate internal integration                             |
| Boundary-conditioned field comparator                    | Established spectral/boundary physics                                                      | Supplies a literal P-layer testbed for context-conditioned admissibility while preserving ordinary-physics ownership                                                                                                                                                                                                          | Low novelty; strong comparator                            |
| Casimir/dynamical-boundary energy gate                   | Established Casimir/QFT/thermodynamic physics                                              | Prevents boundary/vacuum analogies from silently becoming self-power or free-energy claims                                                                                                                                                                                                                                    | Low novelty; important hygiene                            |
| Holographic deformation comparator                       | Established optics                                                                         | Requires explicit encoding/readout/deformation response before calling another system physically holographic                                                                                                                                                                                                                  | Low novelty; useful comparator                            |
| Cross-domain recurrence discipline                       | Structural analogy/convergence established                                                 | Requires typed relation translation and deformation rather than visual resemblance                                                                                                                                                                                                                                            | Moderate methodological role                              |
| SIPO capstone / addressed law assembly                   | Constrained domains, kernels, operator domains, instruments and GR constraints established | Composes <code>Update_SIPO</code> as addressed state/context $\rightarrow$ admissibility descriptor $\rightarrow$ P-law object $(D,T,W) \rightarrow$ P propagation $\rightarrow$ typed $P \rightarrow (P,O)$ instrument $\rightarrow$ readdressing, while retaining changing-address/scale/history guards and the P-only null | Moderate candidate synthesis                              |
| Dynamic-interface promotion / recursive boundary closure | Stateful interfaces, boundary dynamics, junctions and coupled-system variables established | Promotes an interface from bundled context to explicit recursive state only when it is trackable, prospectively load-bearing, recursively updated and—at higher scale—passes existing object-promotion tests; explicitly separates state promotion from ontology promotion                                                    | Low ingredients; moderate candidate operational synthesis |

The confidence labels concern novelty relative to the literature searched in this audit, not truth, importance, patentability, empirical validation, or global historical priority.



## 4. Strongest candidate contribution after the expanded audit

The audit does **not** support saying that Module 32, its supplements, Module 33 or Module 33S1 invented constraint closure, changing possibility spaces, local-to-global consistency, hypergraphs, sheaves, fibers, quotients, multiple realisation, coarse-graining, macro causal emergence, causal semantics, information synergy, holography, parallel transport, holonomy, geometric phase, Casimir physics, constrained dynamics, self-adjoint quantum domains, quantum instruments, ADM constraints, dynamic interfaces, junction conditions, Einstein–Rosen bridges, or vacuum amplification.

The narrower candidate is the following operational conjunction:

*an addressed adaptive formalism in which construction/state-space change triggers explicit comparison guards; current organisation changes admissibility and reachable-state geometry; lower-scale feasible/addressable states are kept separate from higher-scale capability; proposed load-bearing relations are tested by ablation and carrier substitution; identity change triggers separate parity and temporal-continuity tests; typed relation scopes select a compatible assignment region; a missing address is tested as a conditional completion fiber; a higher-scale readout defines macro realisation classes and a quotient but earns endogenous object status only through separate promotion tests; intrinsic relation class is separated from extrinsic contextual address; transport/path dependence is tested explicitly when endpoint equality may hide history; compatible P-realisation is kept distinct from relational specification and actual occupancy; candidate  $I \rightarrow P$  support is allowed to be P-address-changing without false comparison; boundary/holographic physical comparators are deformation-tested; persistent mismatch remains distinct from cost; vacuum/boundary effects remain subject to complete energy accounting; the resulting dependencies are composed into a typed recursive update in which complete addressed state/context produces an admissibility descriptor and active P-law object  $(D, T, W)$  before physical propagation and typed  $P \rightarrow (P, O)$  registration; and a relation/interface is promoted from bundled context to explicit recursive state only when it is operationally trackable, prospectively changes a load-bearing downstream object, is itself changed by realised dynamics, and then changes the next-cycle admissibility, with higher-scale internalisation allowed only after the existing scale-promotion tests pass.*

No single pre-existing source located in this audit supplies that full operational conjunction in the same form.

It is recorded as a **candidate original formal synthesis and operationalisation by Mark Charles McLaughlin**, with each refinement timestamped by its actual public commit on 15 August 2026 and subject to reduction if earlier equivalent work is located.

This wording is deliberately narrower than new law, new mathematics, new theory of life, first theory of relational self, proof of an information layer, proof of holographic reality, wormhole mechanism, quantum-gravity derivation, or vacuum-energy mechanism.

## 5. Why the scale-separated assay matters

The intuition that restriction of components can enable a larger whole is not new. The testable refinement is to keep the observables separate.

Where before/after comparison is lawful:

$$\Delta\mu_\ell < 0, \quad \Delta\nu_L > 0$$

is an enabling-constraint pattern, while

$$\Delta\mu_\ell < 0, \quad \Delta\nu_L < 0$$

is an overconstraint control.

Restriction is not itself progress. Higher-scale capability has to be demonstrated separately.

## 6. Why the address, identity, completion, realisation, transport, law-object, and interface-state guards matter

The current family carries seven distinct anti-flattening rules.

1. **Changing construction address:** if the active state space changes, direct subtraction/equality is not licensed without a translation, embedding, quotient, or common observable.
2. **Changing individuality address:** if constitutive boundary, environment coupling, maintenance or identity-equivalence rules change, the same self lost freedom is not licensed without a continuity map/test.
3. **Path-dependent individuality:** if identity depends on recent trajectory, endpoint similarity is insufficient; the history must be transported and its minimum load-bearing horizon tested.
4. **Missing or coarse-grained structure:** absence is not zero structure and a quotient is not automatically a new physical object. Completion must be computed from the relations touching the missing address, while macro readout persistence is represented by a realisation class and still requires separate effective-object promotion.
5. **Intrinsic versus extrinsic address:** preserved internal relation does not imply unchanged contextual embedding, and changed context does not automatically destroy the internal relation class. When the path through context can be load-bearing, same endpoint is insufficient; compare transport/loop history. Compatible P-realisation remains distinct from actual occupancy.
6. **Admissibility versus law versus registration:** the admissibility descriptor, active P-law components (D,T,W), physical propagation, and  $P \rightarrow (P,O)$  measurement/registration instrument are separate typed objects. Domain-only conditioning is a clean special case rather than a universal mechanism; a changed domain is not automatically a new force, and an observer record cannot stand in for missing P-dynamics.
7. **Interface state versus ontology:** a boundary/relation may deserve explicit recursively updated state representation because it is trackable and prospectively load-bearing even when it is fully recoverable from ordinary P-state variables. State promotion therefore does not imply a new layer, new microscopic degree of freedom, or independent  $I \rightarrow P$  dynamics. Conversely, a visually salient interface remains bundled context when explicit promotion adds no prospective discrimination.

These rules do not assert that environment, history, relation scopes, completion fibers, transport connections, boundary modes, interface states or I-layer variables are always constitutive. Each must earn load-bearing status in the claim being tested.

## 7. Chronology boundary

The first public Module 32 formulation remains:

```
8e4fe4160ba57874c5cf6a7213d44ba3d0c97287
15 August 2026, 01:20:48 BST
```

Later same-day work produced Module 32A, 32S, 32S1, 32S2, 32S3, 32S4, Module 33 and Module 33S1. Their later equations and tests are not backdated into the first Module 32 formulation or standalone preprint v0.1.

Priority statements must therefore remain versioned and resolution-specific.

## 8. Exact-conjunction claim boundary

The current audit supports:

- candidate original formal synthesis;
- candidate original operationalisation;
- no equivalent full conjunction located in the primary literature searched on 15 August 2026;
- publicly formulated by Mark Charles McLaughlin in timestamped, versioned repository objects on 15 August 2026.

It does not support:

- first person to describe constraints, changing possibility spaces, emergence or relational identity;
- first use of equivalence classes, fibers, quotients, hypergraphs, sheaves, coarse-graining, ablation, temporal history, causal semantics, transition kernels, fiber bundles, connections, parallel transport, holonomy, holographic transforms, constrained dynamical domains, self-adjoint boundary conditions, quantum instruments, gravitational constraint surfaces, dynamic interfaces, junction conditions, or Einstein–Rosen bridges;
- first discovery that boundaries affect field spectra or Casimir forces;
- proof that a relationally specified missing address is physically occupied;
- proof that a macro equivalence class is automatically an autonomous physical entity;



- proof that a context-indexed realisation set is a physically existing hidden object;
- proof that an explicit interface macrostate is an independent ontology or I-layer object;
- proof of literal wormholes or traversable shortcuts from a geometric comparator;
- proof of extractable zero-point energy or self-power;
- proof that the full conjunction is globally unprecedented;
- proof that the formalism is empirically superior;
- proof of consciousness, life, independent I→P dynamics, survival after death, a completed quantum-gravity theory, or a new fundamental physical law.

## 9. What would upgrade or downgrade the novelty claim

Upgrade only if broader systematic review across theoretical biology, artificial life, control theory, dynamical systems, multiscale modelling, causal emergence, higher-order networks, differential geometry/gauge transport, category/sheaf methods, inverse problems, diachronic identity, path-dependent systems, interface/junction dynamics, quantum measurement theory, constrained/operator-domain dynamics, general-relativistic Hamiltonian constraints, morphogenesis, quantum thermodynamics, Casimir/boundary QFT, holography and philosophy of science fails to locate an equivalent conjunction **and** the formalism demonstrates discriminating empirical value.

Downgrade immediately if earlier work is found containing the same functional conjunction, even under different terminology.

A partial predecessor does not erase the residual contribution; it changes its boundary.

## 10. Primary literature used in this audit

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## 11. Current verdict

The expanded audit narrows the Module 32/33/33S1 family but does not reduce it to rediscovery. The mathematics of fibers, quotients, hypergraphs, sheaves, coarse-graining, multiple realisation, transport, holonomy, constrained domains, process kernels and measurement instruments is prior art; boundary-conditioned field physics, GR constraint surfaces, junction dynamics, Einstein–Rosen bridge geometry and holography are prior physical/mathematical comparators.

The strongest defensible residual is the **integration**: an addressed, scale-separated, deformation-tested framework in which typed relations define admissibility; absence is represented by a conditional completion fiber; lower-level turnover can be tested inside a macrostate realisation class; scale promotion is separated from observer coarse-graining; intrinsic relation is separated from extrinsic contextual address; path transport is tested when endpoint equality can hide history; compatible P-realisation is separated from actual occupancy; identity continuity is separately guarded; mismatch is separated from cost; an  $I \rightarrow P$  claim must survive an adequate addressed physical-state/history null and causal-identification discipline; boundary/vacuum analogies cannot bypass ordinary field theory or energy accounting; the SIPO loop is closed architecturally by mapping complete addressed state/context to an admissibility descriptor, assembling an active P-law object  $(D, T, W)$ , propagating physically, passing the result through a correctly typed  $P \rightarrow (P, O)$  instrument, and recursively readdressing the realised path; and a load-bearing interface is made explicit in that recursive state only when it passes the trackability, prospective-effect and recursive-back-action gate, with higher-scale internalisation kept behind the existing promotion tests.

That residual is significant enough to preserve canonically and investigate empirically, while remaining open to correction by earlier literature, stronger comparison models, failed derivations, or failed experiments.